Day-1: 01.09.2026

Part A - User and System Investigation (10 minutes)

- Determine the username of the account you are currently using.
  whoami

```
┌──(praneetha㉿Praneetha)-[~]
└─$ whoami
praneetha
```

- Display your UID, primary GID, and supplementary groups.
  UID: id -u, GID: id -g, Groups: id -G

```
┌──(praneetha㉿Praneetha)-[~]
└─$ id
uid=1000(praneetha) gid=1000(praneetha) groups=1000(praneetha),4(adm),20(dialout),
etooth),999(lpadmin)
```

```
┌──(praneetha㉿Praneetha)-[~]
└─$ id -u
1000

┌──(praneetha㉿Praneetha)-[~]
└─$ id -g
1000

┌──(praneetha㉿Praneetha)-[~]
└─$ id -G
1000 4 20 24 25 27 29 30 44 46 100 101 102 114 119 968 979 999
```

- Display all users currently logged in to the system.
  who

```
┌──(praneetha㉿Praneetha)-[~]
└─$ who
praneetha seat0        2026-08-31 21:12 (:0)
```

- Determine what activities the currently logged-in users are performing.
  w

```
┌──(praneetha㉿Praneetha)-[~]
└─$ w
 21:23:42 up 12 min,  1 user,  load average: 0.30, 0.27, 0.16
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
praneeth          -                21:12           0.00s  0.01s lightdm --session-child 13 24
```

- Examine /etc/passwd and identify the following information for your current account:

  Username

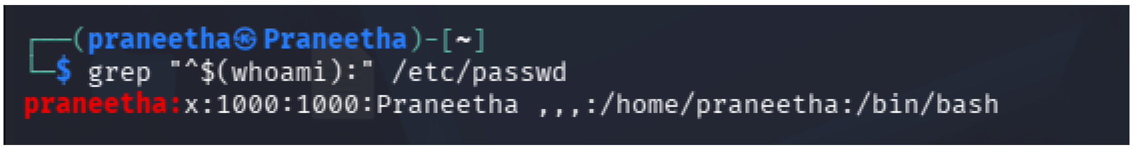
  UID

  GID

  Home directory

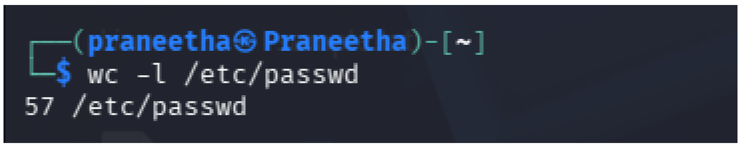
  Login shell

grep “^$(whoami):” /etc/passwd



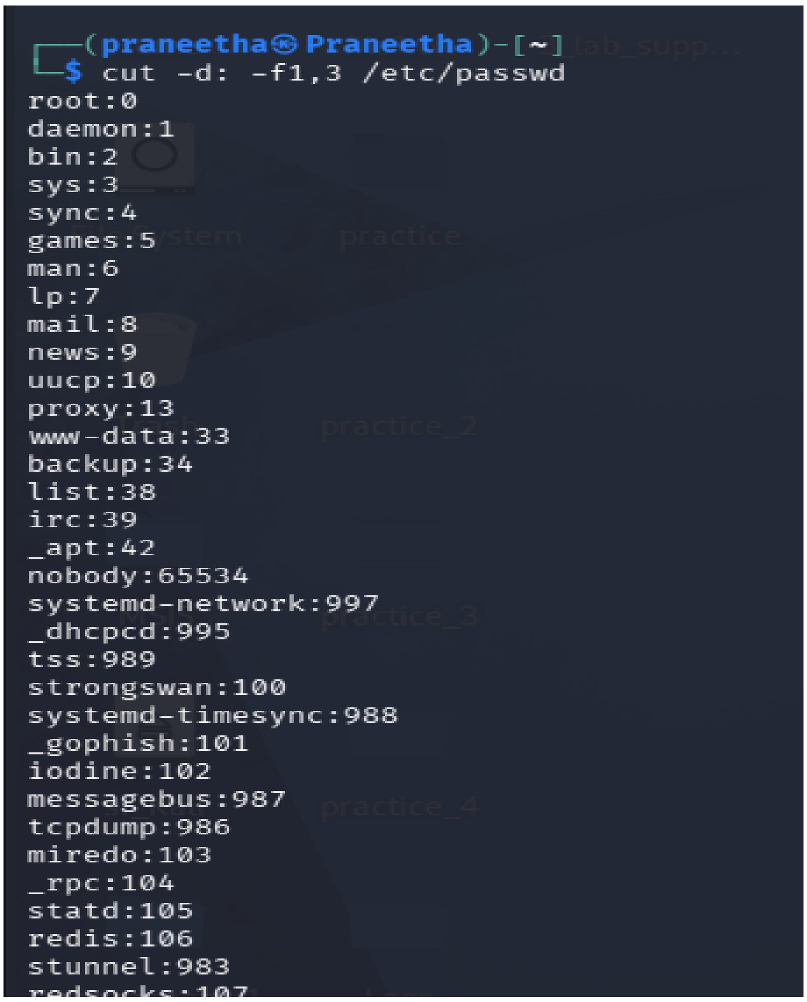
```
┌──(praneetha㉿Praneetha)-[~]
└─$ grep "^$(whoami):" /etc/passwd
praneetha:x:1000:1000:Praneetha ,,,:/home/praneetha:/bin/bash
```

- Determine how many user-account entries are present in /etc/passwd.
  wc -l /etc/passwd



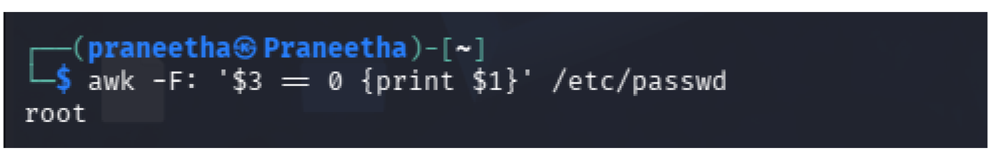
```
┌──(praneetha㉿Praneetha)-[~]
└─$ wc -l /etc/passwd
57 /etc/passwd
```

- Display only the username and UID fields from /etc/passwd.
  cut -d: -f1,3 /etc/passwd



```
┌──(praneetha㉿Praneetha)-[~]
└─$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
irc:39
_apt:42
nobody:65534
systemd-network:997
_dhcpcd:995
tss:989
strongswan:100
systemd-timesync:988
_gophish:101
iodine:102
messagebus:987
tcpdump:986
miredo:103
_rpc:104
statd:105
redis:106
stunnel:983
redsocks:107
```

- Find all accounts whose UID is 0. Explain the security significance of UID 0.
  awk -F: '$3 == 0 {print $1}' /etc/passwd



```
┌──(praneetha㉿Praneetha)-[~]
└─$ awk -F: '$3 == 0 {print $1}' /etc/passwd
root
```

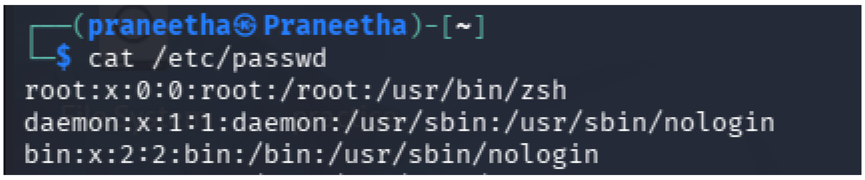
Security significance of UID 0:

UID 0 represents the root/superuser account.

A user with UID 0 has essentially unrestricted access to the system.

An unauthorized UID 0 account is a serious security risk. Attackers who obtain UID 0 effectively have administrator-level control.

- Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?



```
┌──(praneetha㉿Praneetha)-[~]
└─$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
```

/etc/passwd	/etc/shadow
Stores basic information about user accounts.	Stores password hashes and password-related information.
Contains username, UID, GID, home directory, and login shell.	Contains encrypted/hashed passwords and password-aging information.
It is generally readable by all users.	It has restricted permissions and is normally accessible only by root or authorized processes.
The password field usually contains x, indicating that the password hash is stored in /etc/shadow.	Contains the actual password hash.

Password hashes are stored in /etc/shadow because password hashes are sensitive information. The /etc/passwd file needs to be readable by ordinary users and system programs for user and account information. If password hashes were stored there, any user could access them and potentially perform offline password-cracking attacks.

Part B - Creating and Verifying Users (20 Minutes)

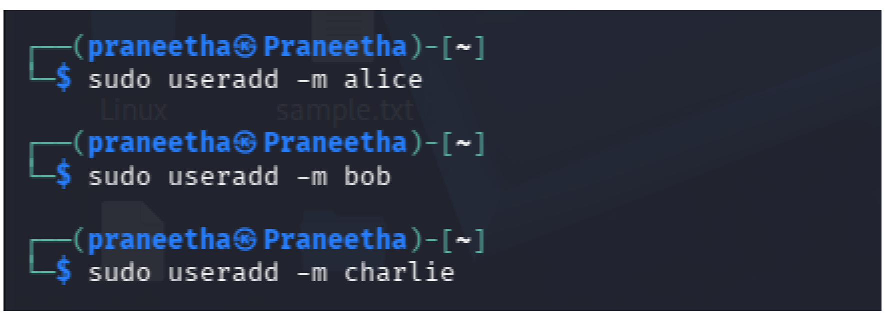
- Create the following users and ensure that a home directory is created for each:

  alice, bob, Charlie

  sudo useradd -m alice

  sudo useradd -m bob

  sudo useradd -m charlie

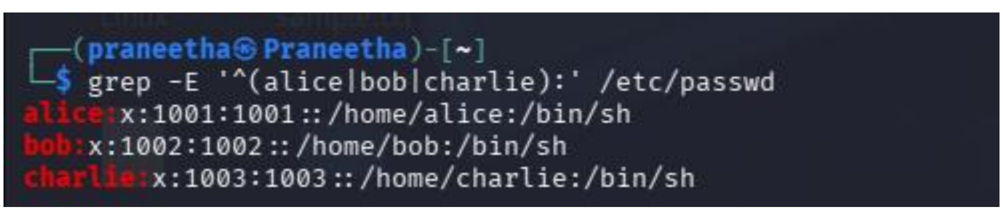


```
┌──(praneetha㉿Praneetha)-[~]
└─$ sudo useradd -m alice

┌──(praneetha㉿Praneetha)-[~]
└─$ sudo useradd -m bob

┌──(praneetha㉿Praneetha)-[~]
└─$ sudo useradd -m charlie
```

- Verify that all three accounts have been added to /etc/passwd.
  grep -E '^(alice|bob|Charlie):' /etc/passwd



```
┌──(praneetha㉿Praneetha)-[~]
└─$ grep -E '^(alice|bob|charlie):' /etc/passwd
alice:x:1001:1001::/home/alice:/bin/sh
bob:x:1002:1002::/home/bob:/bin/sh
charlie:x:1003:1003::/home/charlie:/bin/sh
```

- Determine the UID assigned to each newly created user.
  id -u

```
┌──(praneetha㉿Praneetha)-[~]
└─$ id -u alice
1001

┌──(praneetha㉿Praneetha)-[~]
└─$ id -u bob
1002

┌──(praneetha㉿Praneetha)-[~]
└─$ id -u charlie
1003
```

- Determine the home directory and default shell assigned to alice.
  getent passwd alice

```
┌──(praneetha㉿Praneetha)-[~]
└─$ getent passwd alice
alice:x:1001:1001::/home/alice:/bin/sh
```

Home directory → /home/alice

Default shell → /bin/bash

- Inspect /home and verify that the required home directories were created.
  ls -l /home

```
┌──(praneetha㉿Praneetha)-[~]
└─$ ls -l /home
total 16
drwx------  5 alice     alice     4096 Aug 31 21:41 alice
drwx------  5 bob       bob       4096 Aug 31 21:41 bob
drwx------  5 charlie   charlie   4096 Aug 31 21:41 charlie
drwx------ 18 praneetha praneetha 4096 Aug 31 21:12 praneetha
```

- Set passwords for all three accounts.
  sudo passwd alice

```
┌──(praneetha㉿Praneetha)-[~]
└─$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully

┌──(praneetha㉿Praneetha)-[~]
└─$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully

┌──(praneetha㉿Praneetha)-[~]
└─$ sudo passwd charlie
New password:
Retype new password:
passwd: password updated successfully
```

- Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.
  Switch from your current account to alice: su -alice
  Verify the change: whoami, pwd, id
  return to your original account: exit

```
┌──(praneetha㉿Praneetha)-[~]
└─$ su - alice
Password:
$ whoami
alice
$ pwd
/home/alice
$ id
uid=1001(alice) gid=1001(alice) groups=1001(alice)
$ exit

┌──(praneetha㉿Praneetha)-[~]
└─$ whoami
praneetha
```

Part C - Group Management and Access Control (30 Minutes)

- Create two groups:

  security and developers

  sudo groupadd security

  sudo groupadd developers

```
┌──(praneetha㉿Praneetha)-[~]
└─$ sudo groupadd security

┌──(praneetha㉿Praneetha)-[~]
└─$ sudo groupadd developers
```

- Add alice and charlie to the security group.
  sudo usermod -aG security alice
  sudo usermod -aG security charie

```
┌──(praneetha㉿Praneetha)-[~]
└─$ sudo usermod -aG security alice

┌──(praneetha㉿Praneetha)-[~]
└─$ sudo usermod -aG security charlie
```

- Add bob to the developers group.
  sudo usermod -aG developers bob

```
┌──(praneetha㉿Praneetha)-[~]
└─$ sudo usermod -aG developers bob
```

- Without switching accounts, verify the group membership of all three users.
  id alice or groups alice

```
┌──(praneetha㉿Praneetha)-[~]
└─$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),1004(security)

┌──(praneetha㉿Praneetha)-[~]
└─$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob),1005(developers)

┌──(praneetha㉿Praneetha)-[~]
└─$ id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),1004(security)

┌──(praneetha㉿Praneetha)-[~]
└─$ groups alice
alice : alice security

┌──(praneetha㉿Praneetha)-[~]
└─$ groups bob
bob : bob developers

┌──(praneetha㉿Praneetha)-[~]
└─$ groups charlie
charlie : charlie security
```

- Display the entries corresponding to security and developers from /etc/group.
  grep -E '^(security|developers):' /etc/group

```
┌──(praneetha㉿Praneetha)-[~]
└─$ grep -E '^(security|developers):' /etc/group
security:x:1004:alice,charlie
developers:x:1005:bob
```

- Add bob to the security group without removing him from the developers group.
  sudo usermod -aG security bob

```
┌──(praneetha㉿Praneetha)-[~]
└─$ sudo usermod -aG security bob
```

- Verify that bob now belongs to both groups.
  groups bob

```
┌──(praneetha㉿Praneetha)-[~]
└─$ groups bob
bob : bob security developers

┌──(praneetha㉿Praneetha)-[~]
└─$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob),1004(security),1005(developers)

┌──(praneetha㉿Praneetha)-[~]
└─$ grep -E '^(security|developers):' /etc/group
security:x:1004:alice,charlie,bob
developers:x:1005:bob
```